

Module 2: Electricity and Magnetism

Course content:

1. Static electric and magnetic fields :
We briefly discuss, Coulomb, Gauss and Ampere Laws
Capacitance and Inductance calculations
2. Electrodynamics :
Faradays Law, displacement current, Maxwell's equations
and boundary conditions, electromagnetic wave.

Text books:

1. Introduction to Electrodynamics-- David J Griffiths
2. Engineering Electromagnetics – W H Hayt Jr. et al

Why Study Electrodynamics?

Most essential subject for Scientist and Engineers working in ANY field

What extra you will learn : Detail mathematical treatment of this theory to deal with more complicated real life problems. Deeper understanding for applications.

We will adapt the vector field approach

Scalars and Vectors

Scalars:

Only magnitude, no direction

Example: Distance: 5 m; Speed: 10 m/s; Age: 14 years, etc.

Algebra consists of addition, subtraction, multiplication, etc

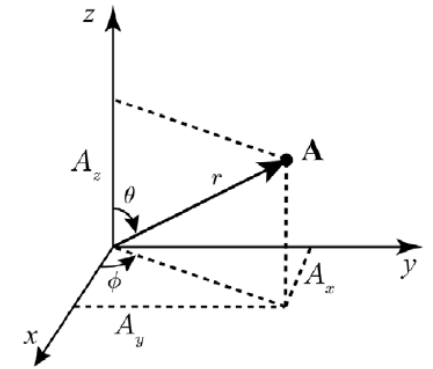
Vectors:

Consists of both magnitude and direction

Requires more than one scalars (components to describe.

Example: Velocity: 5 km/s along the x-direction; $\vec{A} = A_x \hat{x} + A_y \hat{y} + A_z \hat{z}$

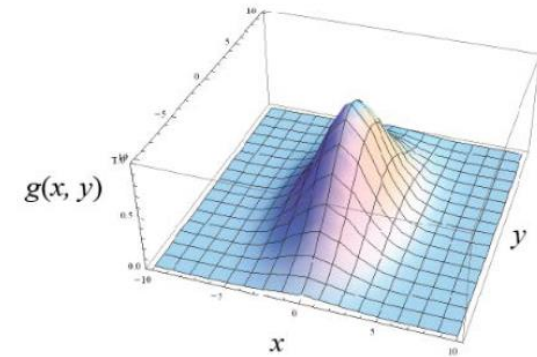
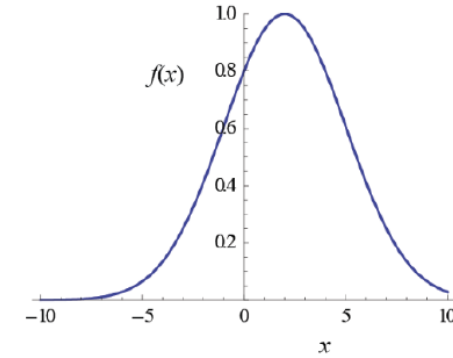
The corresponding algebra is called Vector Algebra



Scalar and Vector functions/fields

Scalar valued functions: A mapping to a scalar field

- Requires only one function for its description
- Example: $f(x)$, $g(x, y)$, $t(x, y, z)$
- Algebra consists of addition, subtraction, multiplication, etc.
- Calculus consists of differentiation, integration, etc.



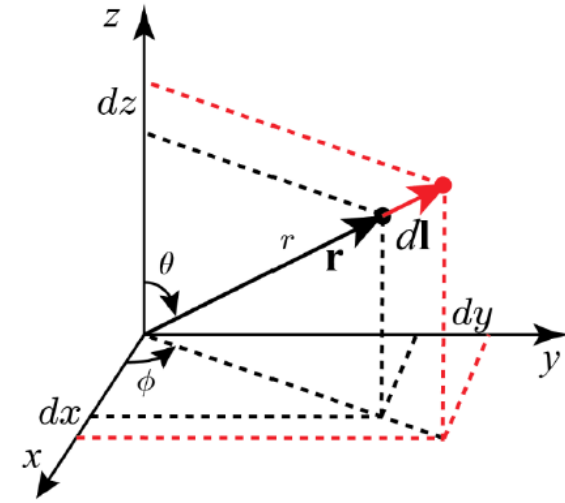
Vector valued functions: A mapping to a vector field

- Requires more than one scalar functions for its description.
- Example: $\mathbf{A}(x, y, z) = A_x(x, y, z)\hat{\mathbf{x}} + A_y(x, y, z)\hat{\mathbf{y}} + A_z(x, y, z)\hat{\mathbf{z}}$
- The corresponding algebra is called Vector Algebra
- The corresponding calculus is called Vector calculus

Examples of vector functions/fields:

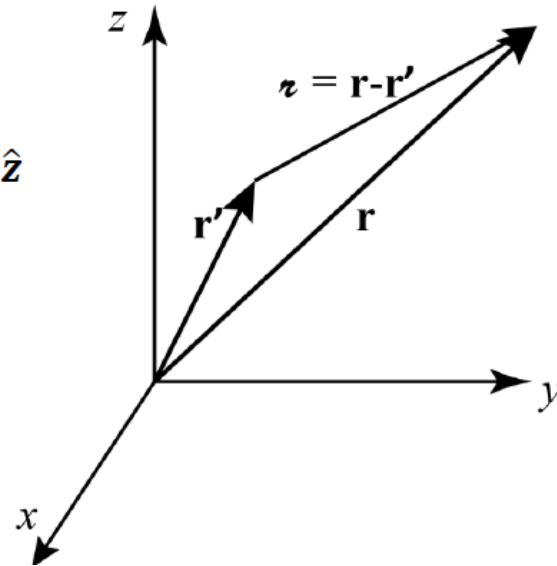
Position vector : $\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$

Displacement vector: $d\mathbf{l} = dx\hat{x} + dy\hat{y} + dz\hat{z}$



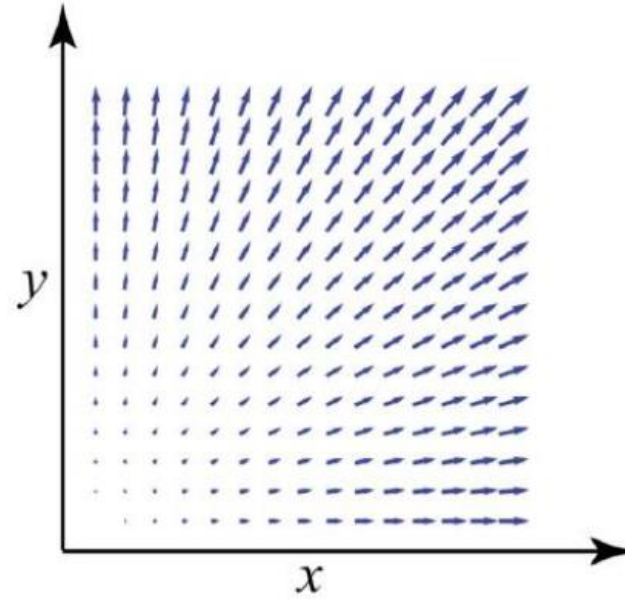
Separation vector:

$$\hat{\mathbf{r}} = \mathbf{r} - \mathbf{r}' = (x - x')\hat{x} + (y - y')\hat{y} + (z - z')\hat{z}$$

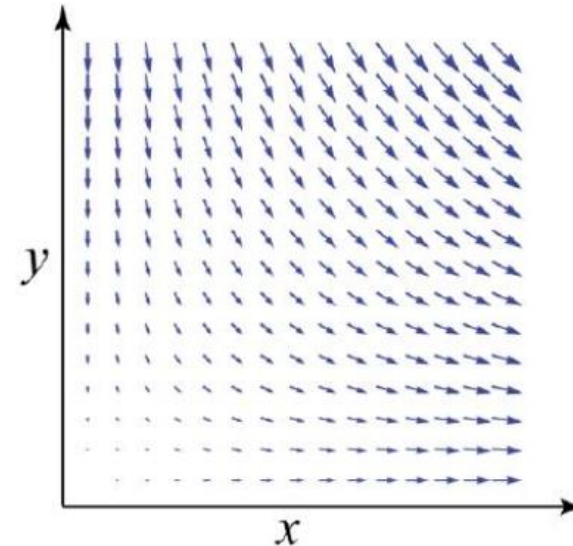


Realization of vector functions/fields:

- Example # 1: $\mathbf{g}(x, y) = x \hat{x} + y \hat{y}$



- Example # 2: $\mathbf{g}(x, y) = x \hat{x} - y \hat{y}$



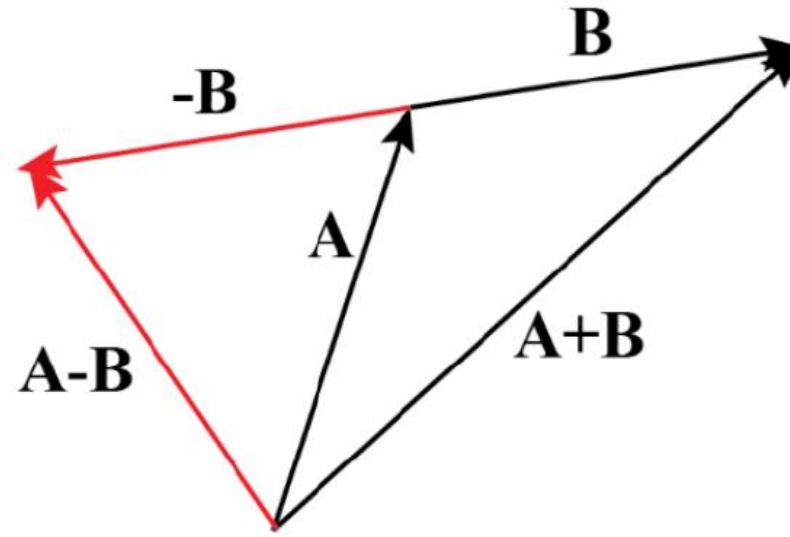
Vector Algebra

Addition:

$$\mathbf{A} + \mathbf{B} = (A_x + B_x)\hat{x} + (A_y + B_y)\hat{y} + (A_z + B_z)\hat{z}$$

Subtraction:

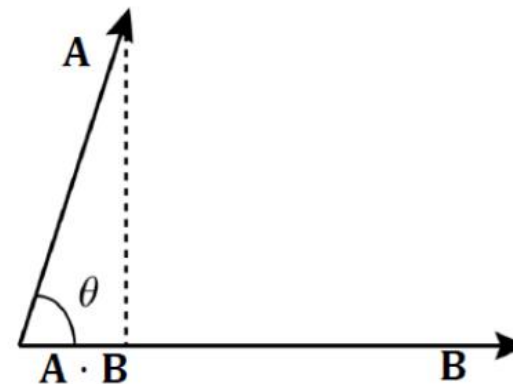
$$\mathbf{A} - \mathbf{B} = (A_x - B_x)\hat{x} + (A_y - B_y)\hat{y} + (A_z - B_z)\hat{z}$$



Multiplication:

(1) Scalar product (dot product):

$$\begin{aligned}\mathbf{A} \cdot \mathbf{B} &= |\mathbf{A}||\mathbf{B}|\cos\theta \\ &= (A_x\hat{x} + A_y\hat{y} + A_z\hat{z}) \cdot (B_x\hat{x} + B_y\hat{y} + B_z\hat{z}) \\ &= A_xB_x + A_yB_y + A_zB_z\end{aligned}$$

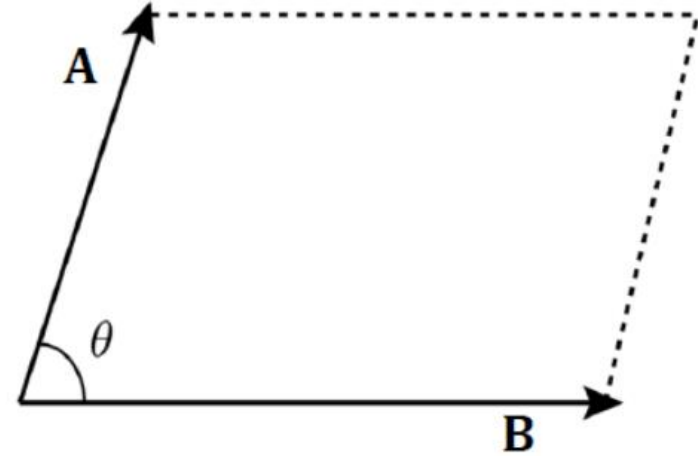


Vector Algebra

(2) Vector product (cross product):

$$\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}||\mathbf{B}|\sin\theta \hat{n}$$

$$= \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} = (A_y B_z - A_z B_y) \hat{x} + (A_z B_x - A_x B_z) \hat{y} + (A_x B_y - A_y B_x) \hat{z}$$



vector product is a vector quantity

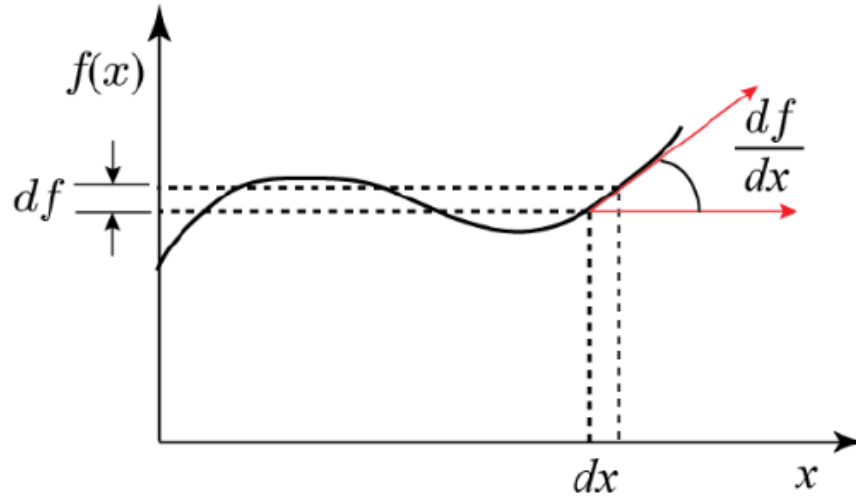
(3) Triple product (Scalar): $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B})$

(4) Triple product (vector): $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$

Vector Calculus : The calculus dealing with vector functions is referred to as the vector calculus. Calculus mainly involves differentiation and integration

Differential Calculus (of function $f(x)$ of single variable x .)

$$df = \left(\frac{df}{dx} \right) dx$$



- The magnitude of $\frac{df}{dx}$ is the rate of change (slope) of function $f(x)$
- If $\frac{df}{dx} = 0$ then $df = 0$, this defines the extremum of function $f(x)$.

Differential Calculus (of function $T(x, y, z)$ of three variables.)

Q: How do we find out the rate of change of a function of more than one variable?

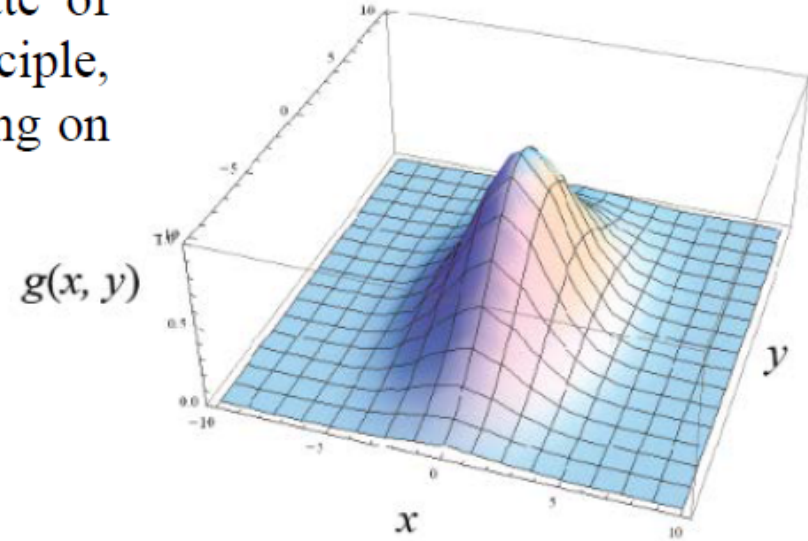
Issue: For a function of more than one variable the rate of change depends on the direction we move. So, in principle, there can be infinite number of “rate of changes” depending on the direction.

Solution:

It is known from a theorem on partial derivatives that a function of three variables can be written as

$$dT = \left(\frac{\partial T}{\partial x} \right) dx + \left(\frac{\partial T}{\partial y} \right) dy + \left(\frac{\partial T}{\partial z} \right) dz$$

- This shows how T changes when the three variables are changed by the infinitesimal amounts dx, dy, dz .
- The above representation means that one does not require an infinite number of “slopes” or “rate of changes.” It is sufficient to know just three.



Differential Calculus (of function $T(x, y, z)$ of three variables.)

Q: How do we find out the rate of change of a function of more than one variable?

Solution: We can rewrite dT as:

$$dT = \left(\frac{\partial T}{\partial x} \hat{\mathbf{x}} + \frac{\partial T}{\partial y} \hat{\mathbf{y}} + \frac{\partial T}{\partial z} \hat{\mathbf{z}} \right) \cdot (dx \hat{\mathbf{x}} + dy \hat{\mathbf{y}} + dz \hat{\mathbf{z}})$$

$$dT = (\nabla T) \cdot (d\mathbf{l})$$

Change in T 3-dimensional derivative displacement vector

- $\nabla T \equiv \frac{\partial T}{\partial x} \hat{\mathbf{x}} + \frac{\partial T}{\partial y} \hat{\mathbf{y}} + \frac{\partial T}{\partial z} \hat{\mathbf{z}}$ is the generalized derivative
- ∇T is called the gradient of T . It is a vector quantity with three components.
- $dT = (\nabla T) \cdot (d\mathbf{l}) = |\nabla T| |d\mathbf{l}| \cos \theta$. For a fixed $|d\mathbf{l}|$, dT is maximum when $\theta = 0$
- So, ∇T points in the direction of maximum increase of the function T .
- The magnitude $|\nabla T|$ is the slope or the rate of change along this maximal direction.
- If $\nabla T = 0$, $dT = 0$. So, $\nabla T = 0$ defines the extremum of the function T .

Del operator and Gradient of a scalar function

$$\nabla T \equiv \frac{\partial T}{\partial x} \hat{x} + \frac{\partial T}{\partial y} \hat{y} + \frac{\partial T}{\partial z} \hat{z} = \left(\frac{\partial}{\partial x} \hat{x} + \frac{\partial}{\partial y} \hat{y} + \frac{\partial}{\partial z} \hat{z} \right) T$$

So, $\nabla \equiv \frac{\partial}{\partial x} \hat{x} + \frac{\partial}{\partial y} \hat{y} + \frac{\partial}{\partial z} \hat{z}$ Is called the Del operator

- ∇ is not a vector in the usual sense, but a vector operator.
- ∇ acts on a scalar function and gives out the generalized derivative

The generalized derivative is call **Gradient** of the scalar function

$\nabla T \cdot \hat{u}$ represents the rate of change of T along the $\hat{u}$ direction called the directional derivative of T along $\hat{u}$.

Example # 1

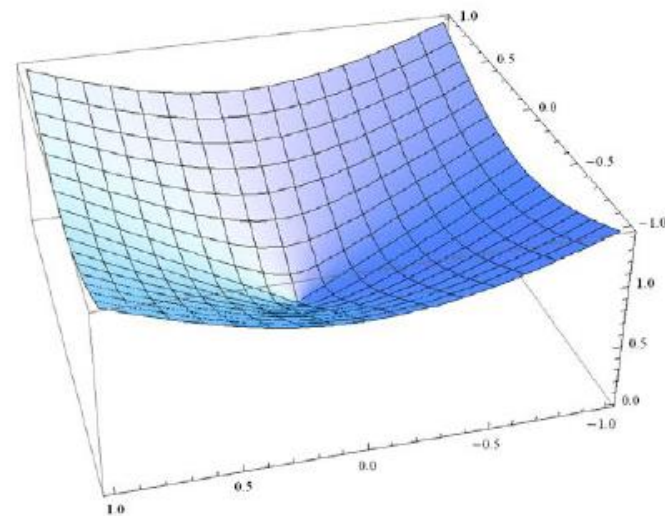
- Find the gradient of $r(x, y) = \sqrt{x^2 + y^2}$

$$\nabla r = \frac{\partial r}{\partial x} \hat{\mathbf{x}} + \frac{\partial r}{\partial y} \hat{\mathbf{y}}$$

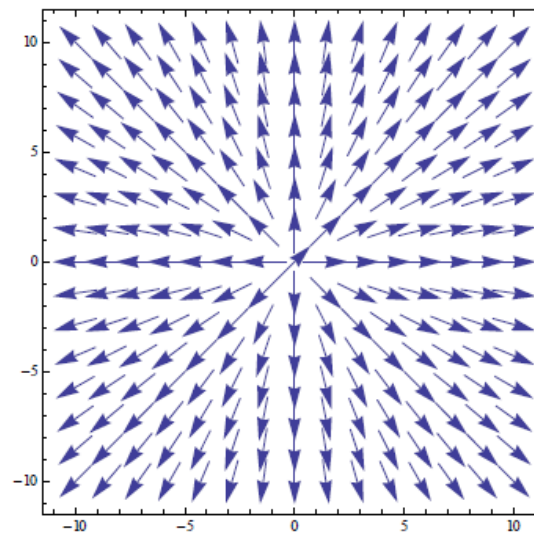
$$= \frac{1}{2} \frac{2x}{\sqrt{x^2 + y^2}} \hat{\mathbf{x}} + \frac{1}{2} \frac{y}{\sqrt{x^2 + y^2}} \hat{\mathbf{y}}$$

$$= \frac{x}{r} \hat{\mathbf{x}} + \frac{y}{r} \hat{\mathbf{y}}$$

$$= \hat{\mathbf{r}}$$



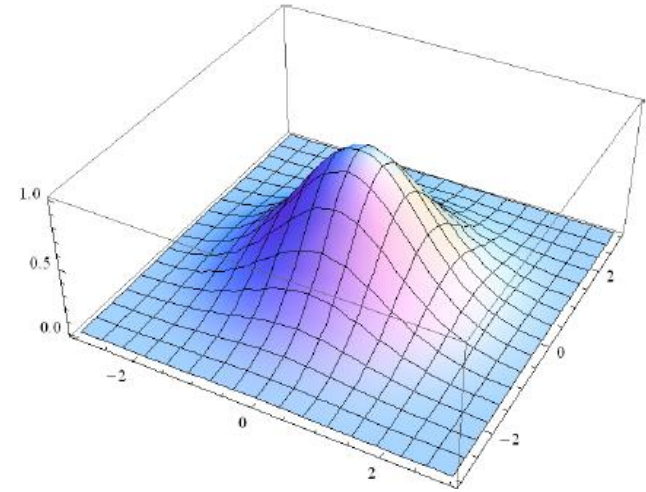
$r(x, y)$



Example # 2

- Find the gradient of $g(x, y) = \exp\left[-\frac{x^2+y^2}{2}\right]$

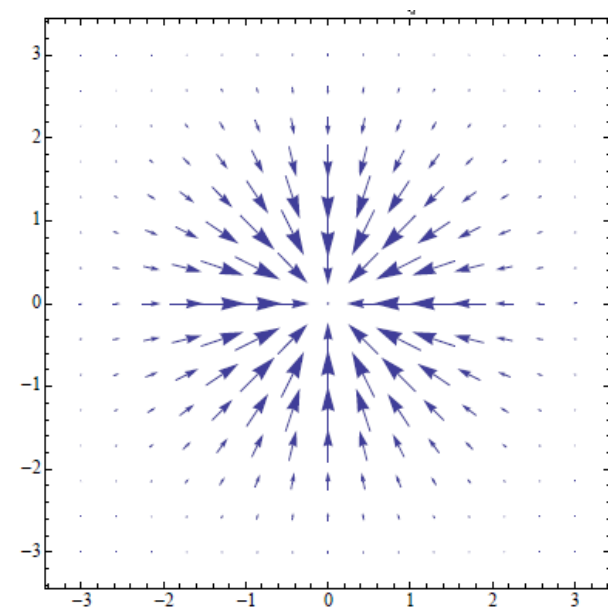
$$\begin{aligned}\nabla g &= \frac{\partial g}{\partial x} \hat{x} + \frac{\partial g}{\partial y} \hat{y} \\ &= -x \exp\left[-\frac{x^2+y^2}{2}\right] \hat{x} - y \exp\left[-\frac{x^2+y^2}{2}\right] \hat{y}\end{aligned}$$



- Find the extremum of $g(x, y)$

$$\nabla g = 0 \quad \text{One solution is } (x, y) = (0, 0)$$

So, the function has an extremum at $(0, 0)$, which is a maximum.



Del operator acting on Vector field

Divergence of a vector $\nabla \cdot \mathbf{V}$

Curl of a vector $\nabla \times \mathbf{V}$

The Divergence

The divergence of a vector $\mathbf{V}$ is defined as

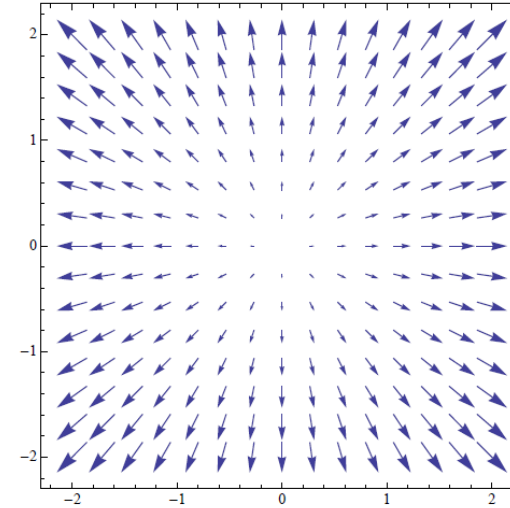
$$\begin{aligned}\nabla \cdot \mathbf{V} &= \left(\frac{\partial}{\partial x} \hat{\mathbf{x}} + \frac{\partial}{\partial y} \hat{\mathbf{y}} + \frac{\partial}{\partial z} \hat{\mathbf{z}} \right) \cdot (v_x \hat{\mathbf{x}} + v_y \hat{\mathbf{y}} + v_z \hat{\mathbf{z}}) \\ &= \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right)\end{aligned}$$

- The divergence of a vector is a scalar quantity.
- The divergence measures how much a vector field diverges.

Example # 1

- Find the divergence of $\mathbf{V} = x \hat{x} + y \hat{y}$

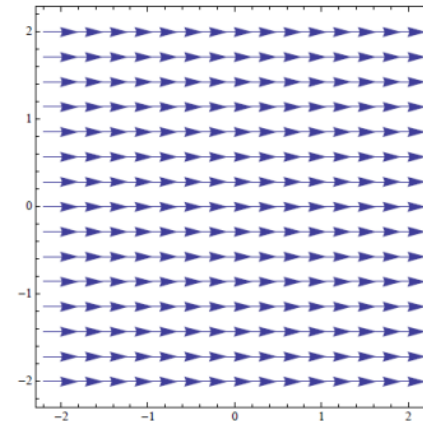
$$\begin{aligned}\nabla \cdot \mathbf{V} &= \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right) \\ &= 1 + 1 = 2\end{aligned}$$



Example # 2

- Find the divergence of $\mathbf{V} = \hat{x}$

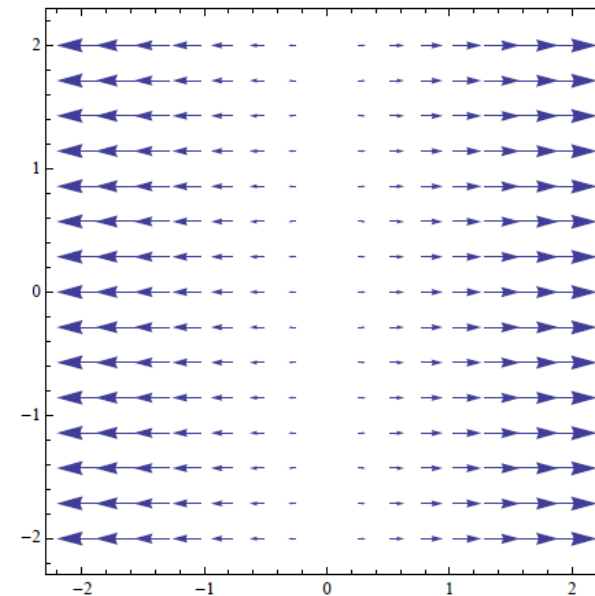
$$\begin{aligned}\nabla \cdot \mathbf{V} &= \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right) \\ &= 0\end{aligned}$$



Example # 3

- Find the divergence of $\mathbf{V} = x \hat{x}$

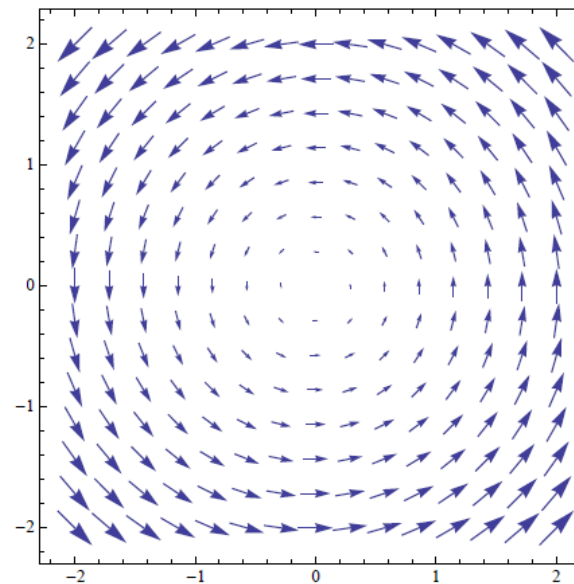
$$\begin{aligned}\nabla \cdot \mathbf{V} &= \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right) \\ &= 1\end{aligned}$$



Example # 4

- Find the divergence of $\mathbf{V} = -y \hat{x} + x \hat{y}$

$$\begin{aligned}\nabla \cdot \mathbf{V} &= \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right) \\ &= 0 + 0 = 0\end{aligned}$$

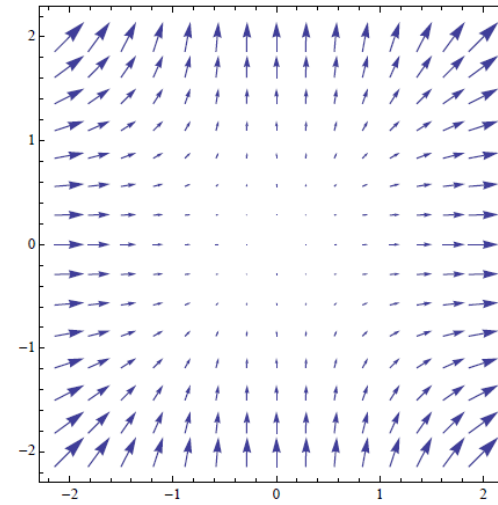


Example # 5

- Find the divergence of $\mathbf{V} = x^2 \hat{x} + y^2 \hat{y}$

$$\nabla \cdot \mathbf{V} = \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right)$$

$$= 2x + 2y$$



The Curl

The curl of a vector $\mathbf{V}$ is defined as

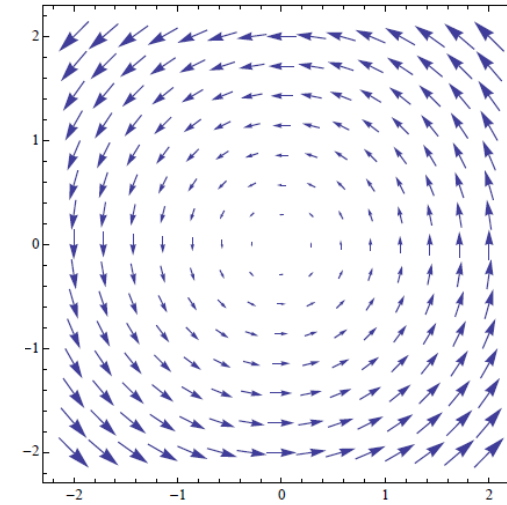
$$\begin{aligned}\nabla \times \mathbf{V} &= \left(\frac{\partial}{\partial x} \hat{\mathbf{x}} + \frac{\partial}{\partial y} \hat{\mathbf{y}} + \frac{\partial}{\partial z} \hat{\mathbf{z}} \right) \times (v_x \hat{\mathbf{x}} + v_y \hat{\mathbf{y}} + v_z \hat{\mathbf{z}}) \\ &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_x & v_y & v_z \end{vmatrix} = \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right) \hat{\mathbf{x}} + \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \right) \hat{\mathbf{y}} + \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right) \hat{\mathbf{z}}\end{aligned}$$

- The curl of a vector is a vector quantity.
- The curl measures how much a vector field curls.

Example # 1

- Find the curl of $\mathbf{V} = -y \hat{x} + x \hat{y}$

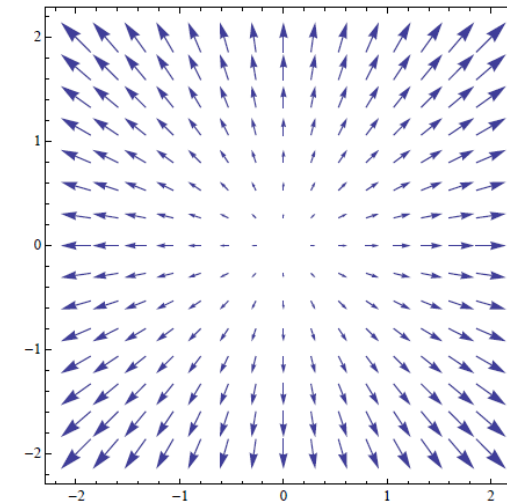
$$\begin{aligned}\nabla \times \mathbf{V} &= \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right) \hat{x} + \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \right) \hat{y} + \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right) \hat{z} \\ &= 2\hat{z}\end{aligned}$$



Example # 2

- Find the curl of $\mathbf{V} = x \hat{x} + y \hat{y}$

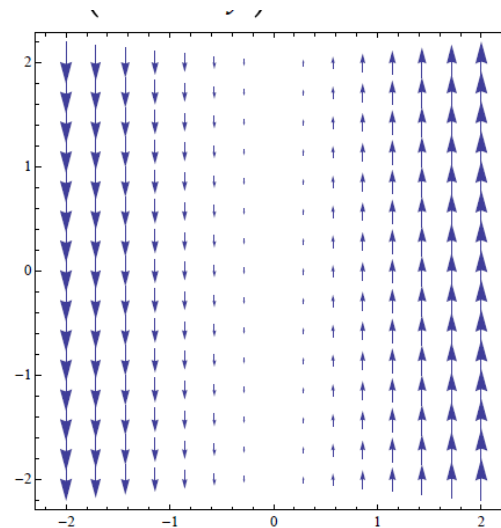
$$\begin{aligned}\nabla \times \mathbf{V} &= \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right) \hat{x} + \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \right) \hat{y} + \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right) \hat{z} \\ &= 0\end{aligned}$$



Example # 3

- Find the curl of $\mathbf{V} = x \hat{\mathbf{y}}$

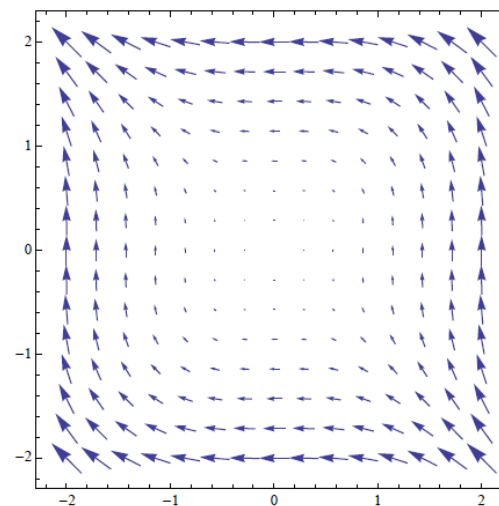
$$\begin{aligned}\nabla \times \mathbf{V} &= \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right) \hat{\mathbf{x}} + \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \right) \hat{\mathbf{y}} + \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right) \hat{\mathbf{z}} \\ &= \hat{\mathbf{z}}\end{aligned}$$



Example # 4

- Find the curl of $\mathbf{V} = -y^2 \hat{\mathbf{x}} + x^2 \hat{\mathbf{y}}$

$$\begin{aligned}\nabla \times \mathbf{V} &= \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right) \hat{\mathbf{x}} + \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \right) \hat{\mathbf{y}} + \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right) \hat{\mathbf{z}} \\ &= 2(x + y) \hat{\mathbf{z}}\end{aligned}$$



Vector calculus with gradient operator

Products Rules:

1. $\nabla(fg) = f(\nabla g) + g(\nabla f)$
2. $\nabla(\mathbf{A} \cdot \mathbf{B}) = \mathbf{A} \times (\nabla \times \mathbf{B}) + \mathbf{B} \times (\nabla \times \mathbf{A}) + (\mathbf{A} \cdot \nabla)\mathbf{B} + (\mathbf{B} \cdot \nabla)\mathbf{A}$
3. $\nabla \cdot (f\mathbf{A}) = f(\nabla \cdot \mathbf{A}) + \mathbf{A} \cdot (\nabla f)$
4. $\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B})$
5. $\nabla \times (f\mathbf{A}) = f(\nabla \times \mathbf{A}) - \mathbf{A} \times (\nabla f)$
6. $\nabla \times (\mathbf{A} \times \mathbf{B}) = (\mathbf{B} \cdot \nabla)\mathbf{A} - (\mathbf{A} \cdot \nabla)\mathbf{B} + \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A})$

Second Derivatives:

1. $\nabla \cdot (\nabla f) = \nabla^2 f$
 2. $\nabla \times (\nabla f) = 0$
 3. $\nabla (\nabla \cdot \mathbf{A})$
 4. $\nabla \cdot (\nabla \times \mathbf{A}) = 0$
 5. $\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$
- $\nabla^2 \equiv \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$
 - ∇^2 is called the Laplacian

Integral Calculus:

The ordinary integral that we know of is of the form: $\int_a^b f(x)dx$

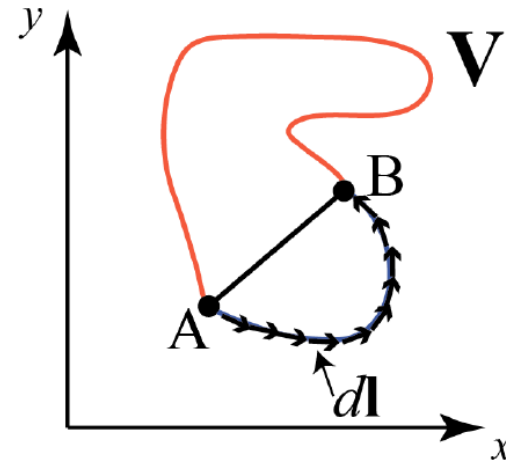
In vector calculus we encounter many other types of integrals.

Line Integral:

$$\int_a^b \mathbf{V} \cdot d\mathbf{l}$$

Vector field Infinitesimal
Displacement vector
Or Line element

$$d\mathbf{l} = dx \hat{\mathbf{x}} + dy \hat{\mathbf{y}} + dz \hat{\mathbf{z}}$$



If the path is a closed loop then the line integral is written as

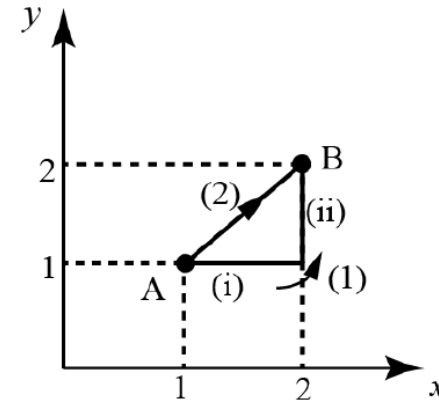
$$\oint \mathbf{V} \cdot d\mathbf{l}$$

- When do we need line integrals?

Work done by a force along a given path involves line integral.

Example (G: Ex. 1.6)

Q: $\mathbf{V} = y^2 \hat{\mathbf{x}} + 2x(y + 1)\hat{\mathbf{y}}$? What is the line integral from A to B along path (1) and (2)?



Along path (1)

- (i) $d\mathbf{l} = dx \hat{\mathbf{x}}; \quad y=1; \quad \int \mathbf{V} \cdot d\mathbf{l} = \int y^2 dx = 1$
- (ii) $d\mathbf{l} = dy \hat{\mathbf{y}}; \quad x=2; \quad \int \mathbf{V} \cdot d\mathbf{l} = \int_1^2 4(y + 1) dy = 10$

Along path (2): $d\mathbf{l} = dx \hat{\mathbf{x}} + dy \hat{\mathbf{y}}; \quad x = y; \quad dx = dy$

$$\int \mathbf{V} \cdot d\mathbf{l} = \int_1^2 (x^2 + 2x^2 + 2x) dx = 10$$

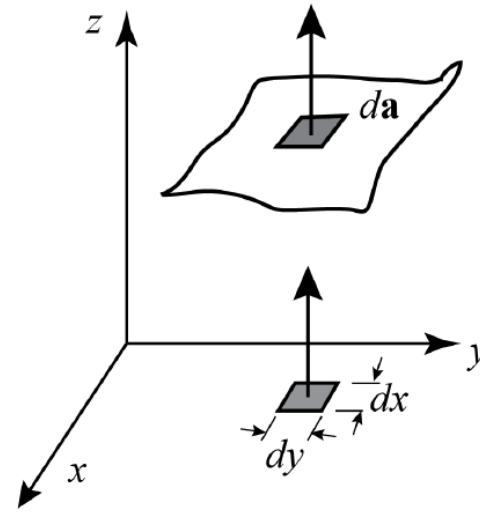
Line integral of the vector field $\mathbf{V}$ from the point A to the point B is path dependent !!!

These vector fields (force-fields) are called **NON-CONSERVATIVE** field

Surface Integral:

$$\int \mathbf{V} \cdot d\mathbf{a}$$

Vector field Infinitesimal
area vector

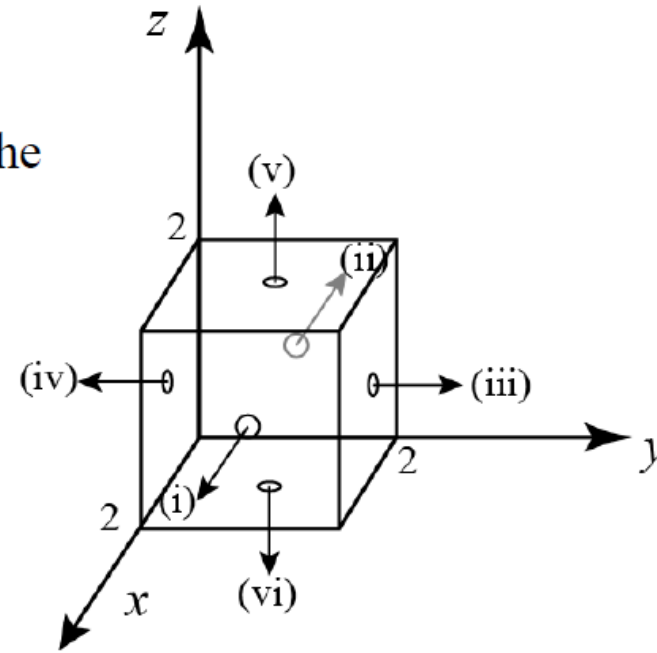


- For a closed surface, the area vector points outwards.
- For open surfaces, the direction of the area vector is decided based on a given problem.
- It represents Flux through a given area
- In the case of closed surface $\oint \mathbf{V} \cdot d\mathbf{a}$ represents total flux coming out of the surface

Example (Griffiths: Ex. 1.7)

Q: $\mathbf{V} = 2xz \hat{\mathbf{x}} + (x + 2)\hat{\mathbf{y}} + y(z^2 - 3)\hat{\mathbf{z}}$? Calculate the Surface integral.

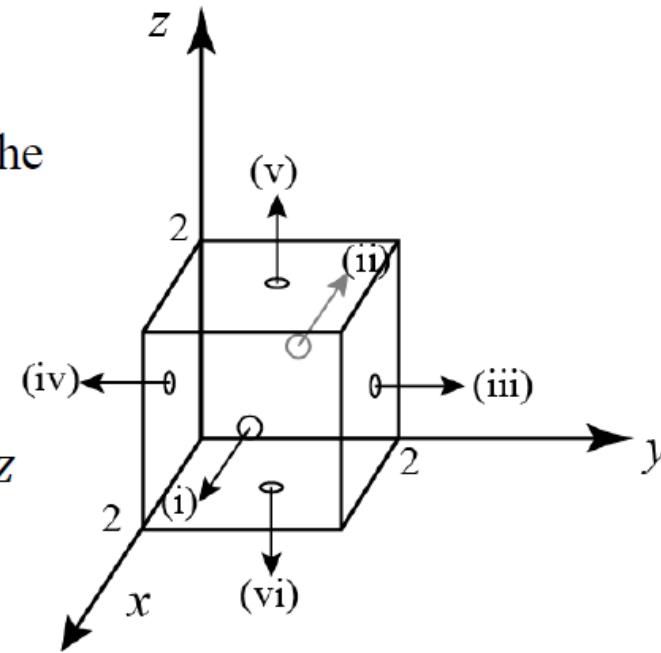
“upward and outward” is the positive direction



Example (Griffiths: Ex. 1.7)

Q: $\mathbf{V} = 2xz \hat{\mathbf{x}} + (x + 2)\hat{\mathbf{y}} + y(z^2 - 3)\hat{\mathbf{z}}$? Calculate the Surface integral.

“upward and outward” is the positive direction



(i) $x=2, d\mathbf{a} = dydz \hat{\mathbf{x}}; \mathbf{V} \cdot d\mathbf{a} = 2xzdydz = 4zdydz$
 $\int \mathbf{V} \cdot d\mathbf{a} = \int_0^2 \int_0^2 4zdydz = 16$

(ii) $\int \mathbf{V} \cdot d\mathbf{a} = 0$

(iii) $\int \mathbf{V} \cdot d\mathbf{a} = 12$

(iv) $\int \mathbf{V} \cdot d\mathbf{a} = -12$

(v) $\int \mathbf{V} \cdot d\mathbf{a} = 4$

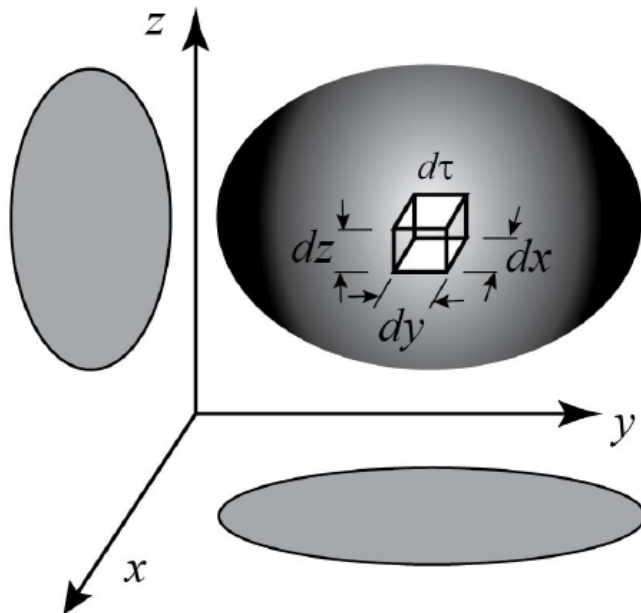
(vi) $\int \mathbf{V} \cdot d\mathbf{a} = -12$

$$\oint \mathbf{V} \cdot d\mathbf{a} = 16 + 0 + 12 - 12 + 4 - 12 = 8$$

Volume Integral:

$$\int T(x, y, z) d\tau$$

- In Cartesian coordinate system the volume element is given by $d\tau = dx dy dz$.
- One can have the volume integral of a vector function which $\mathbf{V}$ as
$$\int \mathbf{V} d\tau = \int V_x d\tau \hat{\mathbf{x}} + \int V_y d\tau \hat{\mathbf{y}} + \int V_z d\tau \hat{\mathbf{z}}$$

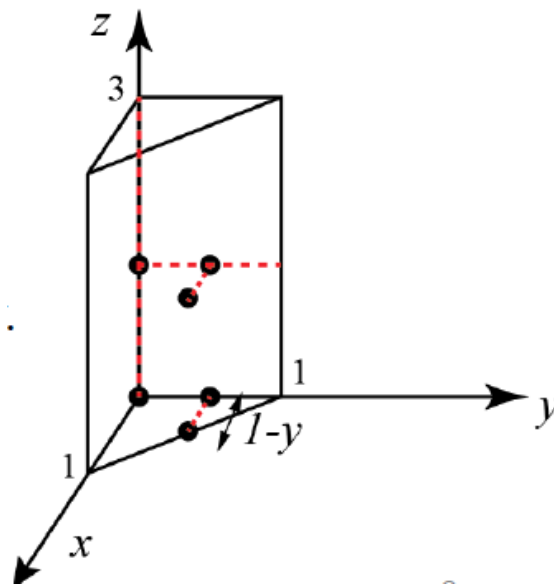


Example (Griffiths: Ex. 1.8)

Q: Calculate the volume integral of $T = xyz^2$ over the volume of the prism

We find that z integral runs from 0 to 3. The y integral runs from 0 to 1, but the x integral runs from 0 to $1 - y$ only. Therefore, the volume integral is given by

$$\int_0^1 y dy \int_0^{1-y} x dx \int_0^3 z^2 dz = \frac{3}{8}$$



The fundamental Theorem of Calculus:

$$\int_a^b F(x)dx = \boxed{\int_a^b \frac{df}{dx} dx = f(b) - f(a)}$$

The integral of a **derivative** over a **region** is given by the difference in values of the function at **boundaries**

Example

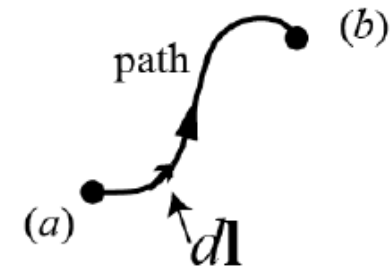
$$\int_a^b x dx = \int_a^b \frac{d\left(\frac{x^2}{2}\right)}{dx} dx = \left[\frac{x^2}{2}\right]_a^b = \frac{b^2 - a^2}{2}$$

- In vector calculus, we have three different types of derivative (gradient, divergence, and curl) and correspondingly three different types of regions and end points.

The fundamental Theorem for Gradient:

$$\int_a^b \nabla T \cdot d\mathbf{l} = T(b) - T(a)$$

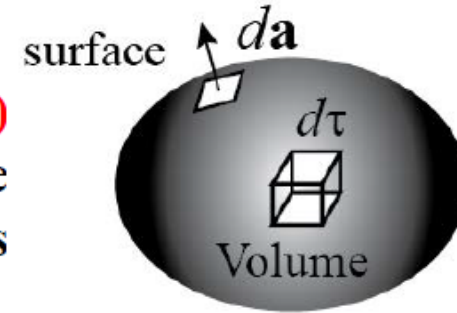
The integral of a **derivative (gradient)** over a **region (path)** is given by the value of the function at the **boundaries (end-points)**



The fundamental Theorem for Divergence (Gauss's theorem):

$$\int_{Vol} (\nabla \cdot \mathbf{V}) d\tau = \oint_{Surf} \mathbf{V} \cdot d\mathbf{a}$$

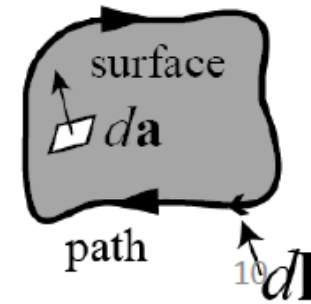
The integral of a **derivative (divergence)** over a **region (volume)** is given by the value of the function at the **boundaries (bounding surface)**



The fundamental Theorem for Curl (Stokes' theorem):

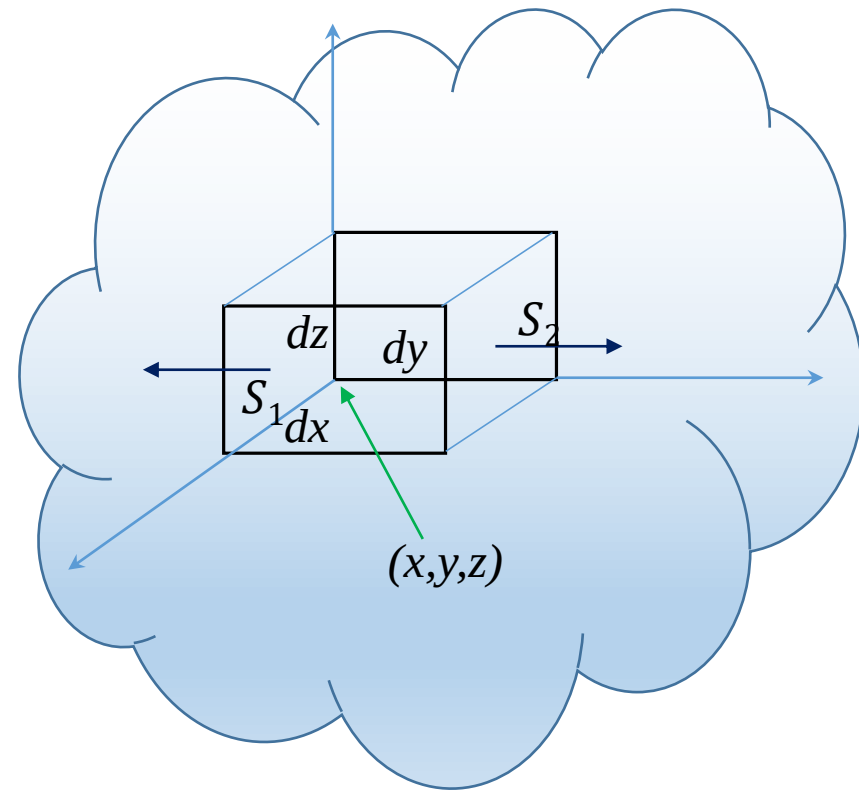
$$\int_{Surf} (\nabla \times \mathbf{V}) \cdot d\mathbf{a} = \oint_{Path} \mathbf{V} \cdot d\mathbf{l}$$

The integral of a **derivative (curl)** over a **region (surface)** is given by the value of the function at the **boundaries (closed-path)**



Gauss's Divergence Theorem

$$\int_{Vol} (\nabla \cdot \mathbf{V}) d\tau = \oint_S \mathbf{V} \cdot d\mathbf{a}$$



Gauss's Divergence Theorem

On the S_1 surface

$$\mathbf{V}(x, y, z) = V_x(x, y, z)\hat{\mathbf{x}} + V_y(x, y, z)\hat{\mathbf{y}} + V_z(x, y, z)\hat{\mathbf{z}}$$

$$d\mathbf{a} = -dxdz \hat{\mathbf{y}} \quad \text{and} \quad \mathbf{V} \cdot d\mathbf{a} = -V_y(x, y, z)dxdz$$

On the S_2 surface

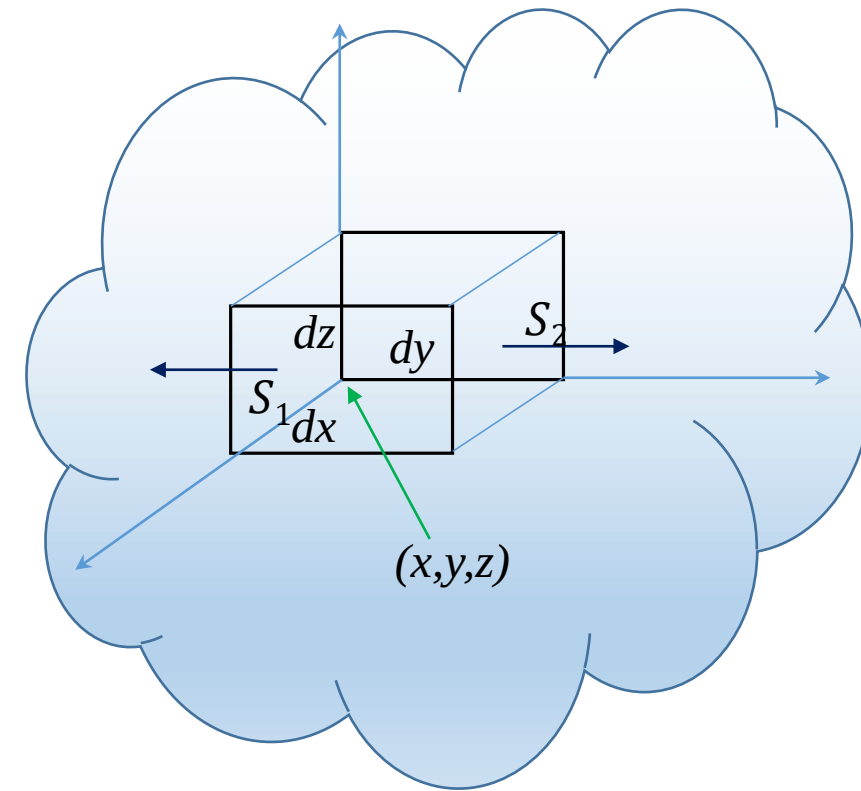
$$\mathbf{V}(x, y + dy, z) = V_x(x, y + dy, z)\hat{\mathbf{x}} + V_y(x, y + dy, z)\hat{\mathbf{y}} + V_z(x, y + dy, z)\hat{\mathbf{z}}$$

$$\begin{aligned} d\mathbf{a} &= dxdz \hat{\mathbf{y}} \quad \text{and} \quad \mathbf{V} \cdot d\mathbf{a} = V_y(x, y + dy, z)dxdz \\ &= \left(V_y(x, y, z) + \frac{\partial V_y}{\partial y} dy \right) dxdz \end{aligned}$$

$$\text{Total surface integral along y-direction} = \frac{\partial V_y}{\partial y} dxdydz$$

Add all infinitesimal close surface

$$\oint_S \mathbf{V} \cdot d\mathbf{a} = \int_{Vol} \left(\frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z} \right) dxdydz = \int_{Vol} (\nabla \cdot \mathbf{V}) d\tau$$

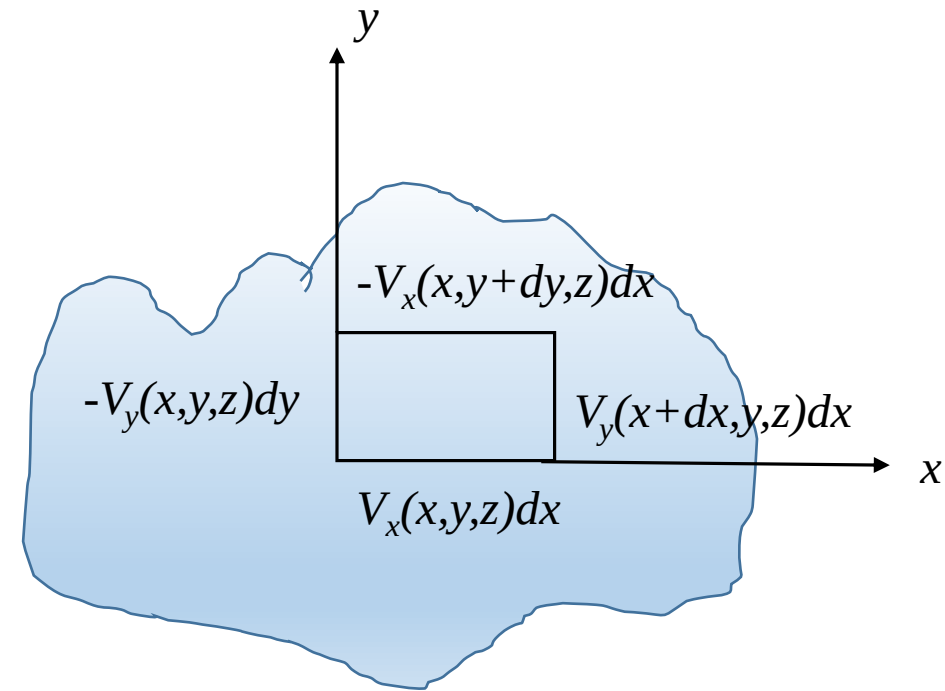


Similarly if we add components along x-, and z-direction, the total surface integral over the closed infinitesimal surface $= \left(\frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z} \right) dxdydz$

Stock's Theorem

Starting with line integral around an infinitesimal loop and adding up we can show

$$\oint_{path} \mathbf{V} \cdot d\mathbf{l} = \int_S (\nabla \times \mathbf{V}) \cdot d\mathbf{a}$$



■

Stock's Theorem

Starting with line integral around an infinitesimal loop and adding up we can show $\oint_{\text{path}} \mathbf{V} \cdot d\mathbf{l} = \int_S (\nabla \times \mathbf{V}) \cdot d\mathbf{a}$

The z-component of the line integral

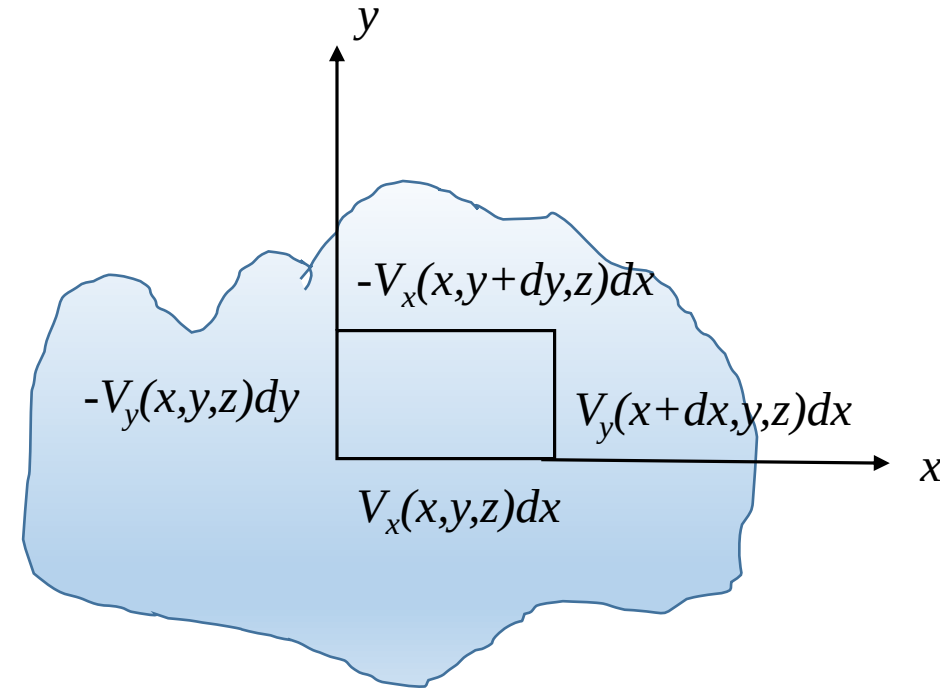
$$\begin{aligned} &= [V_y(x+dx, y, z) - V_y(x, y, z)] - [V_x(x, y+dy, z) - V_x(x, y, z)] \\ &= \left(\frac{\partial V_y}{\partial x} - \frac{\partial V_x}{\partial y} \right) dx dy \end{aligned}$$

Similarly x- and y-component of the line integral are

$$\left(\frac{\partial V_z}{\partial y} - \frac{\partial V_y}{\partial z} \right) dy dz \quad \text{and} \quad \left(\frac{\partial V_x}{\partial z} - \frac{\partial V_z}{\partial x} \right) dx dz$$

Adding all the infinitesimal loops we get

$$\oint_{\text{path}} \mathbf{V} \cdot d\mathbf{l} = \int_{\text{Surface}} \left[\left(\frac{\partial V_z}{\partial y} - \frac{\partial V_y}{\partial z} \right) dy dz + \left(\frac{\partial V_x}{\partial z} - \frac{\partial V_z}{\partial x} \right) dx dz + \left(\frac{\partial V_y}{\partial x} - \frac{\partial V_x}{\partial y} \right) dx dy \right] = \int_S (\nabla \times \mathbf{V}) \cdot d\mathbf{a}$$



Curvilinear Coordinates :

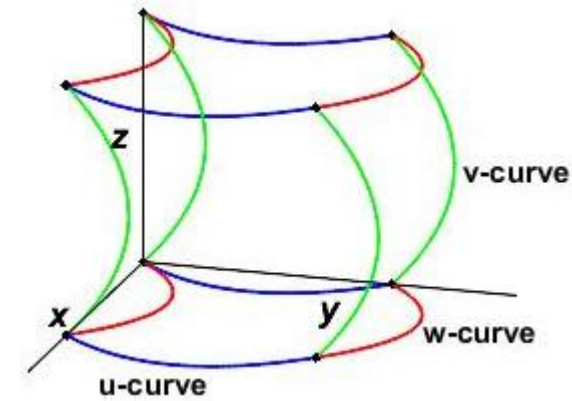
$$(x,y,z) \longrightarrow (u,v,w)$$

$$d\mathbf{l} = dx \hat{x} + dy \hat{y} + dz \hat{z}$$

$$= f(u,v,w) du \hat{u} + g(u,v,w) dv \hat{v} + h(u,v,w) dw \hat{w}$$

$$d\mathbf{A} = g \cdot h dv dw \hat{u} + h \cdot f dw du \hat{v} + f \cdot g du dv \hat{w}$$

$$d\tau = f \cdot g \cdot h du dv dw$$



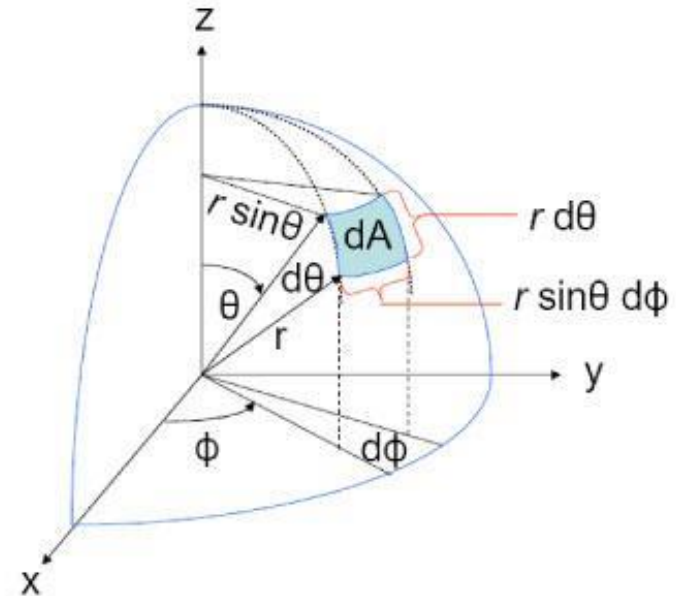
Spherical polar coordinates

$$d\mathbf{l} = dr \hat{r} + r d\theta \hat{\theta} + r \sin \theta d\phi \hat{\phi}$$

$$f = 1, \quad g = r \quad \text{and} \quad h = r \sin \theta$$

$$d\mathbf{A} = r^2 \sin \theta d\theta d\phi \hat{r} + r \sin \theta dr d\phi \hat{\theta} + r dr d\theta \hat{\phi}$$

$$d\tau = r^2 \sin \theta dr d\theta d\phi$$



Curvilinear Coordinates :

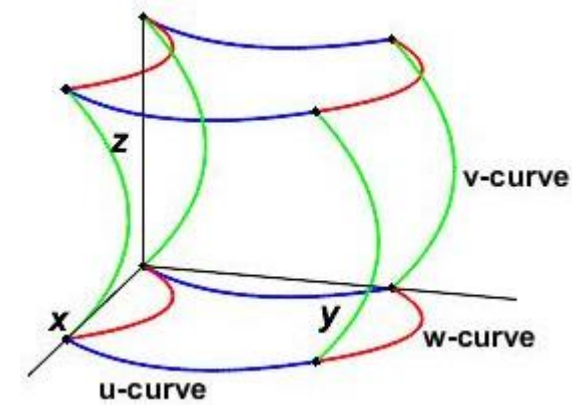
$$(x,y,z) \longrightarrow (u,v,w)$$

$$d\mathbf{l} = dx \hat{x} + dy \hat{y} + dz \hat{z}$$

$$= f(u,v,w) du \hat{u} + g(u,v,w) dv \hat{v} + h(u,v,w) dw \hat{w}$$

$$d\mathbf{A} = g \cdot h dv dw \hat{u} + h \cdot f dw du \hat{v} + f \cdot g du dv \hat{w}$$

$$d\tau = f \cdot g \cdot h du dv dw$$



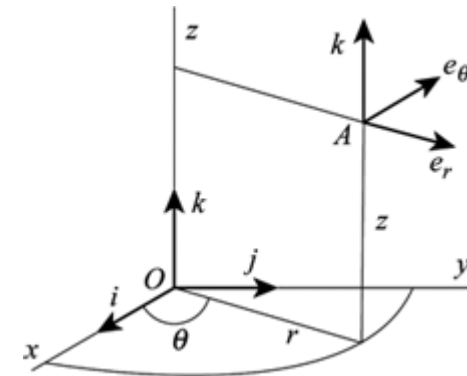
Cylindrical coordinates

$$d\mathbf{l} = dr \hat{r} + r d\theta \hat{\theta} + dz \hat{z}$$

$$f = 1, \quad g = r \quad \text{and} \quad h = 1$$

$$d\mathbf{A} = r d\theta dz \hat{r} + dr dz \hat{\theta} + r dr d\theta \hat{z}$$

$$d\tau = r dr d\theta dz$$



Gradient in Curvilinear Coordinates :

If $\Phi(u, v, w)$ is a scalar function,

$$d\Phi = \frac{\partial \Phi}{\partial u} du + \frac{\partial \Phi}{\partial v} dv + \frac{\partial \Phi}{\partial w} dw$$

$$= \frac{1}{f} \frac{\partial \Phi}{\partial u} f du + \frac{1}{g} \frac{\partial \Phi}{\partial v} g dv + \frac{1}{h} \frac{\partial \Phi}{\partial w} h dw$$

$$\nabla \equiv \frac{1}{f} \frac{\partial}{\partial u} \hat{u} + \frac{1}{g} \frac{\partial}{\partial v} \hat{v} + \frac{1}{h} \frac{\partial}{\partial w} \hat{w}$$

$$= \left(\frac{1}{f} \frac{\partial \Phi}{\partial u} \hat{u} + \frac{1}{g} \frac{\partial \Phi}{\partial v} \hat{v} + \frac{1}{h} \frac{\partial \Phi}{\partial w} \hat{w} \right) \cdot (f du \hat{u} + g dv \hat{v} + h dw \hat{w})$$

$$= (\nabla \Phi) \cdot d\mathbf{l}$$

In spherical polar coordinates

$$\nabla \equiv \frac{\partial}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial}{\partial \theta} \hat{\theta} + \frac{1}{r \sin \theta} \frac{\partial}{\partial \varphi} \hat{\varphi}$$

In cylindrical coordinates

$$\nabla \equiv \frac{\partial}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial}{\partial \theta} \hat{\theta} + \frac{\partial}{\partial z} \hat{z}$$

Divergence in Curvilinear Coordinates :

It can be obtain from
$$\int_{Vol} (\nabla \cdot \mathbf{V}) d\tau = \oint_S \mathbf{V} \cdot d\mathbf{a}$$

